

Preprocessing and NLP Methods

Policy Analysis in the Age of Language Modelling

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TUM Uhrenturm

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Introduction

Overview of today's presentation

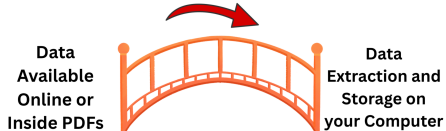


Figure 1 In Workshops 1:3 we have worked on extraction of data available online, or inside PDFs. The goal of these workshop sessions was to give you an idea of the existing and relevant data extraction methods. We move to a new set of pipelines for Workshop 4 and 5: to analyze this data.

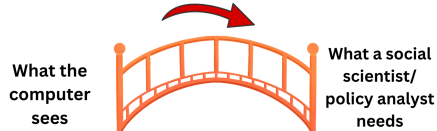


Figure 2 What the computer sees and what we as social scientists/policy analysts need is quite different; both have different ways to "represent" and understand this data. In this workshop, we will look at how to start navigating around this!

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Text Analysis does not assume a single, objective “ground truth.” Instead, it is a dynamic process driven by discovery.

1. No Predefined Outcomes or Methods

- No single “right” answer: multiple valid interpretations depending on your analytical lens.
- Preprocessing is the biggest chunk of analysis

2. Meaning is Social, Not Absolute

- Success is meaningful output onto how humans actually think and communicate.

3. Qualitative Scale with “Power Tools”

- Unlike standard statistical models, text analysis functions like qualitative research.
- Uses NLP algorithms and vector spaces to scale human exploration across many documents.

Current Applications in Text and Content Analysis

Political Science & Policy Analysis

- **Institutional Texts:** Large-scale analysis of parliamentary speeches, legislative drafts, and policy documents.
- **Public Opinion Mining:** Automated tracking of voting behavior discussions, political discourse, and sentiment trends.
- **Multimodal Social Media:** Moving beyond raw text to analyze political memes, video circulation and propaganda, elections and interactions on social platforms.

Digital Humanities

- **Oral Testimonies:** Applying Speech-to-Text and NLP pipeline tools to transcribe, index, and analyze historical audio archives.
- **Literary & Cultural Heritage:** Computational analysis of old books, songs, poetry, and historical language variations to track semantic shifts over centuries.

■ What is a Word?

- ☐ A linguistic label or physical string of characters bounded by spaces.
- ☐ An entry in a dictionary or a label assigned.

■ What is a Concept?

- ☐ Marriam-Webster:
 - something conceived in the mind: thought/notion.
 - an abstract or generic idea generalised from particular instances

■ What is a Document?

- ☐ The primary unit of analysis, ranging wildly in format and scale (e.g., a PDF, a handwritten note, an OCR scan, or a single social media post).
- ☐ Characterized by its encoding, format, length, and structured contents.

■ What is Representation?

- ☐ Spans from structures in the mind/brain to mathematical models in NLP and machine learning.

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Transformation of raw data to make them suitable for analysis

Defined by Substantive Goals

- Done to reduce data complexity *without* altering substantive conclusions existing in the data.
- "Decisions about how words are to be converted into numbers" (Denny and Spirling 2018:168)
- **Domain context guides decisions:**
 - *Policy/Politics*: Certain formal figures of speech must be preserved. Naming conventions, important references etc.
 - *Social Media*: High volume of duplicate quotes/retweets must be filtered out to prevent skewing patterns.

Bag-of-Words Model

The Pipeline Steps

1. Choose the Unit of Analysis

- Define what constitutes a “document” (e.g., speech, paragraph, tweet).

2. Tokenize

3. Reduce Complexity & Clean

4. Create Document-Feature Matrix (DFM)

- Cast text into numerical matrices ready for statistical/language/classification models.

Step 3: Reduce Complexity

■ Lowercasing and Special Characters

- **Noise Removal:** Stripping punctuation, functional URLs, and HTML tags.

- **Stop Word Removal:** Dropping high-frequency, low-semantic words.

- **Equivalence Classes:** Applying stemming or lemmatization (*running*, *runs* → *run*).

- **Context Dependant Cleaning:** For example setting frequency thresholds or weighting guides etc.

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PA Text Preprocessing For Unsupervised Learning: Why It Matters, When It Misleads, And What To Do About It

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Table 1. Preprocessing steps taken/suggested in recent notable papers that deal with unsupervised learning methods. The cite total is taken from Google Scholar at the time of writing. In the case of Slapin and Proksch (2008), we consulted their Wordfish manual (version 1.3). In the case of Roberts *et al.* (2014), the authors suggest further steps might be appropriate for a given application.

Citation	Steps	Cites
Slapin and Proksch (2008)	P-S-L-N-W	427
Grimmer (2010)	L-P-S-I-W	258
Quinn <i>et al.</i> (2010)	P-L-S-I	275
Grimmer and King (2011)	L-P-S-I	109
Roberts <i>et al.</i> (2014)	P-L-S-W	117

Figure 3

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Justin Grimmer and Brandon M. Stewart

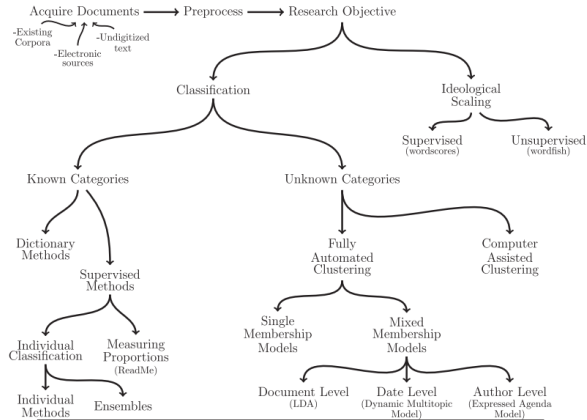


Fig. 1 An overview of text as data methods.

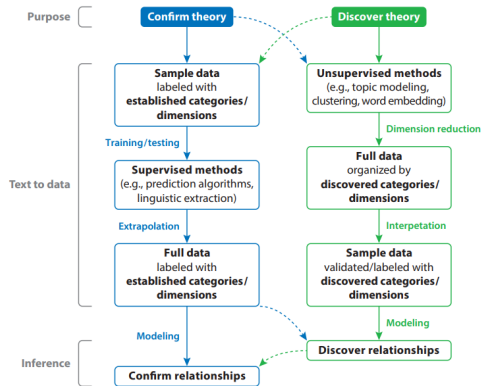


Figure 2

Text analysis and social theory. The arrows trace the text analysis research pipeline, highlighting how different motivations—for confirming versus discovering theory—influence the choice of ML methods used to construct data from text relevant for sociological inference. Solid lines represent straightforward research pipelines, and dashed lines suggest research pathways that mix research motivations for confirmation and discovery. For example, a researcher might explore novel arrangements of established categories (e.g., sentences tagged with positive sentiment). Moreover, once new patterns are discovered in one corpus of text, they may be tested in another.